SPACE FEATS

» Chandrayaan-3 launches a lander and rover on the moon's south polar region in July 2023

» Mangalyaan, the Mars Orbiter Mission, launched in November 2013, reaches Mars orbit on first attempt

» In September 2023, ISRO launched Aditya L1, India's first solar mission, to study the Sun's outer layers

» In the next wave is Gaganyaan, that aims to send Indian astronauts to orbit around Earth

FROM THE ARCHIVES
Apr. 30, 1984

Cover Story
Saga in Space



Anxiously watching the spacecraft's progress on a colour television in their flat of an apartment complex near Moscow were Sharma's wife Madhu and nine-year-old son Kapil... Inside the spacecraft Sharma felt a huge unseen force squashing his body against the seat. He was experiencing gravity's immense downward pull. By now the Soyuz T-11 was gaining height at an amazing speed of 8.1 km per second. Within five minutes it had climbed 160 km. Then exactly nine minutes after blast-off at a height of 200 km above the earth, the spacecraft was injected into space, and Sharma felt the great load on his chest disappear. Free from the clutches of earth's gravity, he had become India's first man in space.
-Raj Chengappa

DID YOU KNOW?

India invested about Rs 1.10 lakh crore in the space sector in the last decade. This created 4.7 million jobs. The Indian space economy is valued around $8.4 billion (Rs 69,720 crore)

"From here, there are no borders. The whole planet is one. It is difficult to understand why there is so much tension in the world. It looks very peaceful from up here"

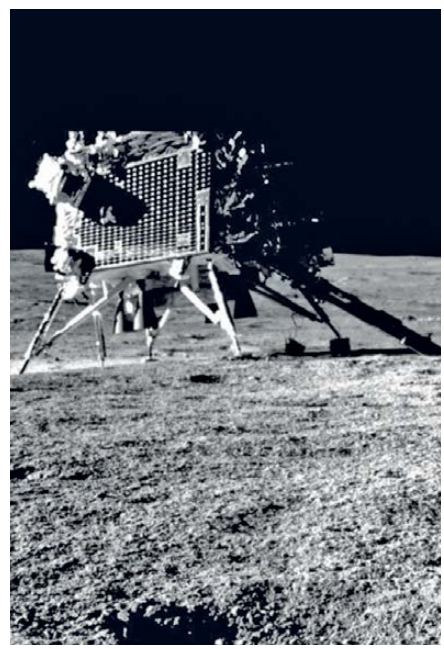
— RAKESH SHARMA

On what he realised as the spacecraft made a close pass over planet Earth

started working on its more ambitious space probes. India joined the space exploration club with the launch of its first mission to the moon—Chandrayaan-1—on October 22, 2008. A PSLV launched the spacecraft, which orbited around the moon to map the lunar surface. Chandrayaan-1 made 3,400 orbits; the ISRO's terrain mapping camera, hyperspectral imager and lunar laser ranging instrument performed to satisfaction. It detected water molecules, particularly in the permanently shadowed craters near the lunar poles. This revolutionised lunar science by challenging the belief that the Moon was dry. Chandrayaan-1 pushed open a door; beyond lay the challenge to explore space in more daring forays.

Chandrayaan-1 was followed by the Mars Orbiter Mission through Mangalyaan, India's first interplanetary spacecraft. Launched on November 5, 2013, the craft carried five payloads to study martian surface features and atmosphere. A craft travelling to Mars cost a paltry Rs 450 crore—this fact fascinated all and raised ISRO's profile. India was the first country to reach the Mars orbit on its first attempt.

Launched in 2019, Chandrayaan-2 failed in its task of landing a rover and a lander on the moon, but advanced India's technical capabilities. Then followed the global headline-making success of Chandrayaan-3, which was launched on July 14, 2023. Its Vikram lander and Pragyan rover followed the codes when Vikram soft-landed in the moon's south polar region—the farthest south that any lunar probe has touched down. Today, India's status as a top spacefaring nation lies on the foundation of Chandrayaan-1's success. It inspired a new generation of scientists, engineers and dreamers. ■



A SMOOTH TOUCHDOWN
Tha Vikram lander on the lunar surface clicked by the navigation camera on Pragyan rover, Aug. 23, 2023